CARVIS

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COMPUTER ENGINEERING DEPARTMENT

CENG 408

Conclusion & Future Work

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Content

1. Contributions

The CARVIS project provides a full-stack clinical AI platform that integrates deep learning-based medical image segmentation with an interactive, physician-facing application. The contributions cover two closely coupled layers, the segmentation pipeline and the application layer that shows segmentation results to end users.

1.1 Segmentation Pipeline

Multi-Organ, Multi-Framework nnUNet Integration

Three independent segmentation modules were designed, implemented and validated, each targeting a different anatomy of clinical interest and using a framework tuned to the data characteristics of that anatomy:

- Brain tumors: nnUNet v2, BraTS 2020 dataset, 5-fold ensemble over the folds 0-4, providing sub-region predictions of enhancing tumor (ET), non-enhancing core / necrosis (NCR/NET) and peritumoral edema (ED).
- Liver parenchyma and tumors: nnUNet v1 (Task003_Liver), 5-fold ensemble, Labels: background, liver, liver tumor.
- Lung tumors: nnUNet v2, Medical Segmentation Decathlon Task06, single-fold checkpoint.

Due to the incompatibility of the Python environments of nnUNet v1 and v2, a dual-environment architecture was adopted, where the liver module spawns an isolated subprocess under a dedicated virtual environment, while the brain and lung modules operate via the Python API within the main application environment.

BraTS Sub-Region Label Normalization

In the BraTS labeling scheme, values 1, 2, and 4 correspond to non-enhancing core/necrosis (NCR/NET), peritumoral edema (ED), and enhancing tumor (ET), respectively. Label 3 is intentionally unused in the BraTS convention, and the pipeline explicitly discards any voxels carrying this value to prevent contamination of downstream metrics. During metric computation, reporting, and uncertainty estimation, all numeric labels are mapped to their clinical names so that the sparse, non-contiguous label space cannot be silently misinterpreted as a dense integer range.

PCA-Based Longest-Axis Diameter Measurement

Rather than determining the diameter from the bounding-box edge (that tends to overestimate non-spherical lesions), the pipeline determines the length of the lesion's projection on the first principal component (PC1) of the point cloud in millimeters. Bounding-box calculation acts as a fallback procedure if the lesion consists of fewer than three voxels or if the covariance matrix is singular.

Harvard-Oxford Atlas Localization

Each lesion is anatomically labeled by aligning the patient's image to the MNI152 1mm template using the SimpleITK rigid and affine transformation, with histogram matching and the Mattes mutual information metric. The new centroid of the lesion in the aligned image is used for querying the Harvard-Oxford cortical-lateralized and subcortical atlases for assigning the lobe, hemisphere, and gyrus labels, along with a registration quality score; the normalized patient's voxel coordinates serve as the last-resort option if alignment or atlas lookup fails.

Dice-Validated Confidence Intervals

The volumetric confidence intervals are calculated based on the internally validated Dice scores rather than applying the uniform ±3% band; for each tissue type, the uncertainty coefficient is set as max(1-Dice, 0.03):

- Enhancing tumor: Dice 0.87, uncertainty ±13%
- Non-enhancing tumor: Dice 0.80, uncertainty ±20%
- Edema: Dice 0.82, uncertainty ±18%
- Lung tumor: Dice 0.65, uncertainty ±35%
- Liver parenchyma: Dice 0.95, uncertainty ±5%
- Liver tumor: Dice 0.60, uncertainty ±40%

Previously, a hardcoded ±3% band was applied to all label types, resulting in significant underestimation of model variance by an order of magnitude for highly variable labels like edema and lung lesions.

Non-ASCII Path Resilience

On Windows systems, where the localization is configured to Turkish, non-ASCII characters are included in the default name of folders (e.g., "Masaüstü" for the Desktop); this causes the NIfTI file writer to silently terminate without providing any feedback. To ensure operation irrespective of the OS localization or OneDrive configuration, a tiered approach was designed: (i) if the path is ASCII-safe, keep it; (ii) otherwise, derive the legacy short path from the system to retrieve the ASCII path; (iii) finally, write the file into the temporary directory and then move it to its destination path.

1.2 Application Layer

FastAPI REST Backend

The backend serves the following structured REST API interfaces, via six domain-scoped routers: cases, CT cases, viewer, report, Q&A, and upload. All API endpoints serialize/deserialize and validate request and response bodies using Pydantic schemas, and each API is automatically documented with OpenAPI via FastAPI, easing external integration efforts in the future.

Intent-Driven Clinical Q&A Engine

A multi-layer natural language question and answering engine takes free text clinical questions about a loaded case. The pipeline applies (1) a spell correction layer using a domain-enriched medical vocabulary; (2) a local intent classification against a typed intent tree with intents for quantitative, spatial, comparative, prognostic and management domains; (3) optional LLM-based intent arbitration for ambiguous questions, and (4) session memory to carry resolved intents across turns in order to facilitate multi-turn conversations.

Automated Structured Report Generation

The report generator is clinician-facing structured reports with six required sections: Clinical Context, Findings, Impression, Limitations, Suggested Clinical Correlation, and Disclaimer; and nine Findings sub-sections: volumetric analysis, spatial localization, confidence scores, VR-ready status, mass effect, midline shift, pattern of enhancement, and multifocality. Every report includes an LLM-generated narrative when it exists, with rule-based alternatives if it does not.

Clinical Safety Guardrails

A specific guardrails module implements three safety constraints on all generated outputs: (1) recognition and blocking of definitive diagnostic terms (e.g., "This is definitely glioblastoma"), (2) generation of low-confidence outputs triggers a required, low-evidence safety note, (3) all clinical outputs fall under one or more rule sets for blacklisted semantic structures before being shipped to the client.

Electron Desktop Application & One-Click Deployment

A desktop application based on Electron encapsulates FastAPI API and React user interface to provide an entirely offline, physician-facing Graphical user interface. Short, automated scripts ensure the virtual environment exists, all dependencies are installed, optionally fire up an LLM server, automate a health-check poll, and make the initial first-time install on a new clinical PC a single double click and no configuration. Uses a separate, first-run setup script to automate the full dependency install process for a non-Python environment.

Immersive VR Volume-Rendering Module (Meta Quest 3)

In addition to the Electron desktop client described above, a standalone Meta Quest 3 application (MedDemo) is shipped that takes the same NIfTI outputs from the desktop pipeline and renders them as fully interactable 3D volumes inside a head-mounted display. Built in Unity 6 (6000.3.7f1) on the Universal Render Pipeline, this application includes the Meta XR SDK 85.0.0 to provide native hand tracking and controller input, and uses a heavily forked open source UnityVolumeRendering package as its raycasting implementation. The build is exclusively Quest 3 (ARM64, multiview stereo, 90 Hz) and is sideloaded as a single signed APK.

The VR module makes the following concrete contributions to the wider CARVIS platform:

- **Runtime NIfTI ingestion without rebuild:** A new bootstrap component (RuntimeVolumeBootstrap) will decompress patient volumes at run time using a 3-step path resolution (explicit absolute path-> Quest persistent data path->bundled StreamingAssets). New cases can be deployed to the headset by adb-pushing a .nii or .nii.gz file into the application persistent data folder; no apk rebuild, app reinstallation or dev machine is needed in between patients.
- **Auto-fitted, hand-graspable volumes:** Whether a volume is grab-able, or "hand-graspable", is determined dynamically during compression. Its size is normalized to the package's bounds-fit primitive, then remains thus rescaled so the world-space length of the longest edge is about 0.35 m an empirically determined size to fit the natural arm span of the user. Two independent grab affordances (VolumeGrab, a grabable grid-aligned volume container with radius 0.28 m, and a sphere "handle" for the cross section tool with radius 0.12 m) are attached.1 A static co-ordinator prevents both grab-able features from being caught at the same time redundant input in the most common nested grab hierarchy cases.
- **Dual-input interaction model:** VolumeGrab can accept Touch controllers (grip / trigger) and bare-hand pinch gestures via OVRHand. The same scene can be used by clinicians in either input mode without recompilation (since the hand-tracking path is superimposed on the controller path). This is particularly useful for sterile or peri-op review.

- **Object-driven cross-sectioning:** A spherical handle is the combined location of a CrossSectionPlane (clipping the volume) and a SlicingPlane (rendering the cut as a 2D MPR-style image on a world-space quad). A single rig controller (RuntimePlaneRigController) drives the two planes so that the user only has to interact with a single"object, rather than two coupled gizmos. To prevent the volume from blinking out of existence as the application loads, the cross-section is gated by a movement threshold (ActivateCrossSectionOnSphereMove): volume clipping doesn't activate until the sphere has moved a tiny distance, ensuring that the user always starts out seeing the full volume.
- **Clinical transfer-function presets with controller toggle:** Three 3D Slicer-derived presets are provided -- CT-Lung, CT-Cardiac3, and CT-Chest-Contrast-Enhanced, encompassing most common clinical scenarios of thoracic CT; specifically, chest, cardiac, and contrast-enhanced thorax CT. Clicking the Quest right-controller B button toggles between each preset in the order of Default -> CT-Lung -> CT-Cardiac3 -> CT-Chest-CE -> Default. The toggle uses the New Input System action-binding API with a polling-based XR fallback that keeps the toggle working on devices where action-map activation is deferred.
- **Quest 3 GPU and shader-stripping countermeasures:** A mobile-target shader optimizer (Quest3ShaderOptimizer) reduces the volume sampling rate to 0.65 and disables tricubic filtering on Adreno-class GPUs, maintaining the point of rendering at a higher refresh rate reprojection while still remaining inside the headset thermal envelope. Three runtime-instantiated shaders (the URP volume shader, the slicing-plane shader, and the custom 2D HUD shader) are explicitly registered in the project's Always-Included Shaders list to ensure the build pipeline does not strip them from the build when none of the scene assets reference them at compile time.

2. Lessons Learned

CARVIS development revealed many engineering and scientific lessons regarding clinical AI systems that are universally applicable to them.

2.1 Segmentation & Model Integration

- **Lesson 1: Compatibility issues with Framework version is a first class deployment concern.**

Running nnUNet v1 and nnUNet v2 side-by-side in the same python environment results in quiet import errors and incorrect results. Running one framework per virtual environment, and having them talk through subprocess creates a sane, extendable solution.

- **Lesson 2: Fixed estimates of uncertainty can be deceptive to the clinician.**

Initial design employed a fixed +/-3% confidence band across all tissue labels. Internal calibration showed this under-represented the 'true' segmentation uncertainty by anywhere up to 6x for high-variance tissues (brain edema: Dice 0.82 -> true error 18%). Anchoring of the confidence band to statistically validated Dice scores vastly increased report credence.

- **Lesson 3: The avgs do not tell you what you need to know. Per-case variation needs to come to the fore, not be smoothed away through averaging.**

Lung Tumor Segmentation achieved an average Dice of 0.65, a standard deviation of 0.27, and a median of 0.79. These statistics showed a highly bimodal distribution. Showing only the average flattenes the distribution because the system performs terribly on some cases. Confidence (High/Moderate/Low) labels were added to inform end users of this variability.

- **Lesson 4: Small tumors of liver: a still open issue**

Liver tumor segmentation (Dice 0.60) was consistently the lowest in performance. Many sub-centimeter lesions were not detected which is a common limitation of present day nnUNet training data rather than an inference pipeline. It emphasizes the importance of having to make people aware of model limitations in generated reports.

2.2 System Engineering & Deployment

- **Lesson 5: Filesystem encoding is an invisible killer in production.**

OneDrive paths on Turkish-locale Windows systems have non-ASCII characters (“Masaüstü”, “Kullanıcılar”) that cause SimpleITK NIfTI writer to crash without producing a comprehensible error message. Identifying and avoiding these paths before a data-loss failure can occur is valuable in pragmatic deployment environments.

- **Lesson 6: A offline first architecture is almost a requirement for a clinical environment.**

Hospital environments often control outbound access to the internet. Making the LLM feature a fully optional upgrade (with robust rule based fallbacks) rather than an integral part guaranteed that it would work in many different deployment scenarios.

- **Lesson 7: domain specific NLP preprocessing cannot be done by general libraries.**

It took a significant effort to create a qualifying medical vocabulary extension as otherwise correctly spelled words, such as ;, enhancing, and glioblastoma are marked as errors by standard spell-checks and it is impossible to properly classify the intent of a medical question.

- **Lesson 8: LLM guardrails must be explicitly forbidden-pattern enforced, not just prompted.**

Giving the LLM a system prompt to “avoid definitive diagnoses” mitigates but doesn‘t completely prevent unsafe outputs. The additional post-generation validation layer to compare against a predefined collection of “forbidden patterns” gives a deterministic safety net independent of model behavior.

- **Lesson 9: Mobile XR silently builds strip shader instantiated by running them.**

Volumerendering assets in CARVIS are generated on-the-fly by the application, they are not authored within a scene. Consequently any build pipeline for Unity treats the relative shaders as unreferenced, removing them from the APK. This results in a functioning editor build however it does not render the volume on the headset. A reliable way to overcome this is to add all runtime shader GUIDs in ProjectSettings/GraphicsSettings.asset into m_AlwaysIncludedShaders, although URP shader GUIDs are known to vary and this will need to be re-verified after each package update.

- **Lesson 10: Quest 3 GPU budget is very tight for direct volume rendering.**

A naive port of the desktop volume shader to Quest 3 achieved single digit frame rates and caused Asynchronous SpaceWarp, both of which are unacceptable in a clinical scenario. Three mitigations altogether brought the target refresh rate back to a stable higher value reducing the maximum number of steps, not reducing the mobile sampling rate to 0.65, and disabling tricubic interpolation on Adreno GPUs. URP render scale was also lowered to 0.8 with 4x MSAA, giving a perceptually better trade off than rendering at full resolution without MSAA.

- **Lesson 11: Sideload-friendly data layouts collapse the patient-rollout cost.**

A traditional pipeline would necessitate rebuilding and resigning an APK (using a call such as "android.bat -xf android-debug.apk AndroidDebug.apk") for each additional patient case. First resolving NIfTI paths dynamically to the Quest's persistent data directory eliminates this deployment loop, resulting in a single "adb push case.nii.gz", and restart of the application. This is a (relatively) inexpensive engineering change with outsized clinical influence turning the headset from a static-content demo unit into a versatile, per-patient diagnostic that can be refreshed on the order of seconds by any medical professional.

3. Applications

CARVIS is envisioned as an inclusive clinical AI infrastructure platform. The fundamental ability to abstract highly complex goals into practicable units underpins very tangible application cases for clinical practice, medical education, research, and public outreach. We have developed a showcase website (produced using Next.js14, React Three Fiber, and interactive 3-D organ atlas pages) that introduces the platform not only to users of existing installations but also to clinicians, hospital administrators, and researchers seeking an understanding of the platform before installation.

3.1 Clinical Practice

Radiology Decision Support

Automated calculations of tumor volume, maximum tumor diameters based on PCA measurement, tissue sub-region breakdowns, anatomic location are returned to the radiologists within seconds after cases are uploaded for review on MRI studies. The structured report format with a mandatory limitation/disclaimer section ensures added AI findings are a supporting evidence, not an end diagnosis.

Neurosurgical Planning

The Harvard-Oxford atlas localization module determines if a tumor borders or invades the functionally eloquent cortex (e.g. frontal motor areas, temporal language areas). Location relative to eloquent cortex, estimate of midline shift, and evaluation of mass effect all together can show a pre-operative overview that can be compared to basic imaging in the patient.

Oncology Longitudinal Monitoring

Standardized volumetric metrics with Dice-validated confidence intervals provide inter-visit comparability. The batch processing engine allows serial follow-up scans to be processed as a group, producing individual case-level JSON metrics files that can be imported into a hospital information system or research database.

Resource-Limited and Offline Settings

As the entire system of CARVIS is based on local hardware without any reliance on any cloud infrastructure, it can be implemented in developing countries in district hospitals, mobile imaging vans or even on low bandwidth facilities. Utilization of GPU accelerations (with fallback to CPU when a GPU is not available) also allows the system to operate on regular clinical workstations.

Immersive Pre-Operative Planning in VR

Surgeons also use the Meta Quest 3 client to explore a patient CT or MRI volume in full stereoscopic 3D, move it freely at one-to-one hand scale, and reveal any internal anatomy with a hand manipulated cross-section. When contrasted with review on traditional flat-panel multi-planar reformats, this headset preserves depth perception and supports full bimanual interaction a key advantage in spatially demanding cases such as thoracic and cardiothoracic or craniofacial surgery. Because the volume is loaded from the headset‘s local storage and the app is independent of network connectivity, the workflow is readily deployable directly in the operating room anteroom or bedside both of which regularly lack reliable cloud network connectivity or an available clinical workstation.

3.2 Medical Education

Interactive Case-Based Learning

The Q&A engine repurposes all the stored cases into interactive didactic cases, allowing medical students and radiology residents to learn by asking questions about tumor volume, anatomic site, Enhancement patterns and differential considerations in natural language and receiving rich, well-organized evidence-grounded responses. Memory for sessions facilitates multi-turn pedagogical discussions that shape clinical logic stepwise.

Segmentation Quality Assessment Training

Confidence intervals and model validation metadata are directly accessible to the user, imparting knowledge about the limitations of the automation. It also prepares for future clinicians to question automation, rather than blindly accept it, a skill increasingly expected by medical AI regulation.

VR-Based Anatomy and Volume-Rendering Practice

The same Meta Quest 3 app that allows clinical pre-operative planning is also a nearly-free volumetric anatomy lab for medical students and radiology residents, allowing them to load and freely translate and scale any educational CT or MR dataset with their hands, toggle between all three clinically-relevant transfer functions (lung, cardiac, contrast-enhanced soft tissue) at a single button press, and use the cross-section tool to interactively demonstrate the connection between surface anatomy and internal structures. Since each transfer function preset is associated with a standard 3D Slicer clinical convention, the student is given an intuition for the same windowing used in regular practice, though it is reinforced by the headset‘s stereoscopic depth cues that make volumetric relationships easier to understand than on a flat monitor.

3.3 Research

Batch Cohort Analysis

A batch evaluation harness allows large case cohorts to be fed through a stat generator program that creates standardized metric files for each case that can be used directly for subsequent statistical analysis, model comparison, and cleaning of the dataset.

Model Development Scaffold

The modular segmentation architecture also means that new nnUNet checkpoints can easily be added to CARVIS by adding one Python module and registering a new API path. CARVIS becomes a rapid prototyping environment for clinical AI researchers who want to quickly develop and validate new segmentation models.

3.4 Outreach and Knowledge Translation

CARVIS Showcase Website

The showcase site was created along with the clinical desktop application in order to show other clinicians, hospital administrators, and researchers its functionality to those who may not have the installed desktop tool available to them at their institution. While the desktop version was created in our standard environment of Electron, React and D3, the showcase was created in Next.js14 (App Router, Typescript) using Tailwind CSS. Motion animation was handled in an all-in-one Anime.js library rather than in individual React components via Framer Motion so that all timing and easing remained the same on every page. Modality-specific 3D organ atlas pages (brain, liver, and lung) utilized React Three Fiber allowing users to hover and click on a 3D model to examine a specific organizer within the context of an anatomical region and read treatment-oriented clinical descriptions. The first view of the homepage featured a particle brain, created in Spline, which moved in response to the mouse cursor to give a taste of the neuroimaging focus of the platform at a glance without requiring any special technical knowledge.

There is also a separate page for platform architecture (the technology stack, pipeline summary, system diagram), segmentation execution performance (per-organ Dice results with confidence intervals and all), and a contact page with a feature for contacting the developers through the Resend email API. The pages listed above use content separated into per-organ TypeScript modules in a content/ directory, so that if a new organ for segmentation is added (e.g., prostate or kidney), only a new content module, and a new path registration are necessary; the render layer does not have to change. This separation of concerns in the website architecture reflects the modular approach of the clinical pipeline as well, so that the pipeline and website evolve in parallel as new disciplines, organs, validations, and releases are achieved.

4. Next Steps

The priorities for development presented below have been concluded based on existing system shortcomings, user feedback examples and the overall clinical AI framework.

4.1 Segmentation Enhancements

- Longitudinal alignment and delta reporting: automatically co-register serial scans from the same patient and calculate signed volume change and growth-rate estimates between visits.
- Broader organ coverage: to include kidney, prostate, pancreas and colon, add segmentation modules for them with the existing MSD and nnUNet benchmark datasets.
- Liver tumor model enhancement: the proprietary nnUNet test set Dice of 0.60 for liver tumor is inadequate for clinical applications requiring a high degree of certainty. Additional fine-tuning on more cases or switching to a more current checkpoint of the nnUNet v2 is a short-term objective.
- Uncertainty visualization: show the per-voxel uncertainty as a heat map overlay in the 3D viewer so clinicians can find out spatially where model is most uncertain.
- More extensive validation: validate the model on a larger validation cohort than the n=5 (brain) and n=13 (lung) cases, providing a statistically reliable (Dice) estimate, preferably by tumor grade size, and MRI protocol.

4.2 Application Enhancements

- Native DICOM viewer integration the pipeline at present is NIfTI based. Native DICOM series viewer, can remove the conversion process for the hospital borne studies.
- PACS / HL7 FHIR communications: connection to hospital PACS systems through DICOM SR or HL7 FHIR diagnostic report resources would enable the CARVIS findings to be included in the electronic health record.
- Turkish-language report generation: the current Q&A engine supports Turkish by leveraging translingual intent match. Moving to fully Turkish-language structured reports would make reports easier for in country clinical users.
- Multi-user access and audit logging: before entering multi-physician setting or formal clinical trials, a role-based access control layer with non-modifiable audit logging is needed.
- Larger embedded LLM-the current deployment uses 7B-parameter GGUF models, avoiding demands on standard hardware of clinical workstations. Once GPU memory grows on commodity hardware, an upgrade to 13B-32B models will result in significantly better quality of report narratives and dialogues/Q&A.
- A responsive web interface would enable most bedside case review to be performed on tablets, which represents much of the workflow during ward rounds. This would also be equally usable from mobile phones.
- Multi-volume and segmentation overlay in VR: Currently, the Quest 3 client only displays one grayscale volume at a time. It would be nice if the output mask of the segmentation pipeline could be visualized as a color overlay (using per-label opacity controls) over the source volume(s), providing surgical teams an opportunity to view the structures the AI detected in 3D stereoscopically, pre-operatively.
- Bimanual volume scaling: the current Quest 3 volume client loads each volume scaled to a fixed largest-edge size; it supports translation/rotation via single hand grab. a bimanual pinch-to-scale gesture-a combined two-hands grab that scale the volume linearly based on the inter-hands distance-might allow surgeons to zoom in on small lesions for fine-grained inspection or pull back for an overview of full anatomy within the immersive view without reaching out and grabbing a menu.

4.3 Validation & Regulatory

- Prospective clinical study: a reader study will measure the time-to-report, measurement reproducibility, and diagnostic confidence of radiologists with and without CARVIS assistance.
- Ethics and data governance framework: for clinical deployment, an approved IRB data de-identification plan, de-identification process and model governance documentation in accordance with the applicable MDD regulations (EU MDR, FDA SaMD guidance).

5. Conclusion & Future Work

5.1 Summary

CARVIS is an end-to-end clinical AI system that extends the cutting edge of deep learning segmentation with a ready-for-deployment clinician-facing app. From an initial raw NIfTI upload all the way to receiving a fully structured radiology report with Dice-validated confidence intervals and atlas-based anatomic localization, the entire pipeline runs on an entirely offline workstation, accessible without the reliance on the cloud or high end DevOps skillsets. A related demo website (interactive 3D organ atlas pages, modality-specific performance tables, modular Next.js app architecture) invites clinicians and evaluators to dabble in the system before deploying it for themselves.

The system supports 3 anatomical targets: brain tumors (BraTS sub-regions), liver with tumors, and lung tumors with all 3 validation on public benchmark datasets; the application layer pairs these models with an intent-driven Q&A engine, a guardrail-protected report composition, and a one-click deployment stack that reduces the clinical adoption barrier. In tandem with this analytic/report pipeline, the Meta Quest 3 client wraps the same volumes for stereoscopic, hand-driven inspection on an offline headset, linking the platform from raw image to written report to immersive surgical pre-review.

5.2 Broader Impact

The long term benefits of the proposed CARVIS system transcend the speed advantages of a computer. The designed system, by consistently and reliably providing quantitative, reproducible measurements offers to help hospital staff regarding one of the widely persisting problems of medical imaging of patients- inter-observer variability. It has been established by research that radiologists' manual measurements of tumor volume exhibit large inter- and intra-observer variation and the proposed computerized segmentation with known quantified uncertainty establishes a more trustworthy reference point for treatment response.

For physicians (and prospective physicians) right now, we will reduce the burden of quantitative analysis. For instance, by providing radiology residents the ability to access this Q&A engine while working on a case, and receiving a concise, direct answer that is based directly off of the segmentation data from that case (rather than some standard textbook description), radiology trainees can query the Q&A engine while reviewing a case and receive answers derived from that case's segmentation output. Compared to static reference material, this case-grounded feedback shortens the loop between observation and explanation.

In Resource-Limited Settings, such as regional hospitals, mobile imaging centers, and hospitals in the developing world, the offline-first, 1-click deployment paradigm can make AI-supported diagnostics available to much broader populations that have historically only had access through well-endowed academic medical centers. The lack of cloud subscription, internet connection, or fee-for-inference, removes traditional barriers to adoption at the point of care.

This Meta Quest 3 client, the companion to this piece, further extends this democratization argument from analytic AI into immersive space visualization. Volumetric preoperative review has been limited traditionally to dedicated (or cloud subscription operated) surgical-planning workstations and platforms costing tens of thousands of dollars a seat, limiting primary use to tertiary academic centers. A consumer grade standalone headset with a sideloadable APK drops that cost by roughly two orders of magnitude, removes the cloud dependence, and removes the specialized PC workstations, while the same device can serve as a stereoscopic anatomy laboratory for trainees, collapsing immersive 3D from the acquisition to a flagship-only feature to something that a regional hospital, a residency program in a low-income country, or a combat support hospital can use routinely.

5.3 AI as a Copilot, Not a Replacement

Guiding principles of the CARVIS design are that the system should never replace, but always assist the clinician. To that end, at several levels in the design, these principles are invoked. The guardrails module prevents forbidden diagnostic phrases, while all quantitative measurements are issued with a Dice-verified measure of uncertainty. Structured reports must include a 'Limitations' section and a warning about clinical correlations. Confidence labels (High / Moderate / Low) present a barrier-free, automatic presentation of segmentation confidence without necessitating the user understands Dice.

This policy isn't just a regulatory measure; it is a recognition-based understanding that almost all of today's deep learning approaches, including nnUNet, will sometimes give a failure mode unintuisvely to a non-expert end-user. Open communication of uncertainty is how this device is earned, not taken for granted.

5.4 Future Vision

In the short term, the most valuable features would be the Longitudinal Delta reporting, the PACS connectivity, and a larger organ coverage library. These features would enable CARVIS to evolve from a stand-alone case analysis application into an integrated "module" of the clinical image viewing workflow that could also be used for multi-visit oncology monitoring using the existing hospital IT systems. The Quest 3 client would benefit most from bimanual pinch-to-scale, runtime overlay of the source volume's segmentation mask, and a wrist-anchored hand-menu for transfer-function and case selection, essentially making the headset a self-contained pre-operative inspection station that would provide the AI results on-demand, across all three dimensions, in stereo 3D.

In the medium term, a prospective reader study will be required to quantify the benefit of the system on radiologist efficiency and reproducibility of measurements. This information will be important both for the scientific literature and to support the regulatory pathway necessary to release CARVIS as a certified medical device under the EU MDR or FDA SaMD standards. The addition of the CARVIS Q&A engine via the Quest's in headset microphone will enable surgeons to query the grasped volume by voice and view the response as a HUD overlay. A blinded reader study comparing the headset client to flat panel review will also be necessary to facilitate broader adoption than early adopters.

Long term, this modular structure enables CARVIS to provide the infrastructure of a larger clinical AI ecosystem. Incremental inclusion of new imaging modalities (CT angiography, PET/CT), anatomies, and analytical functionality (radiomics feature extraction, treatment response prediction) becomes straightforward by adapting the pre-existing modules, rather than forcing a reworking of the platform. As LLM technology advances, and as local hardware optimization improves, Q&A and report generation modules stand to benefit and further narrow the gulf between automated analysis and the clinical report written by skilled radiologists today. The Color Passthrough feature of Quest 3 provides a tantalizing gateway toward true mixed-reality utilization within the operating room once the segmented volume is registered to the real patient, it can be visualized on location; and when combined with Meta XR co-presence, the same platform can facilitate remote tumor boards and surgical rehearsals in a shared three-dimensional environment.

5.5 Closing Remarks

The work at CARVIS illustrates that it is possible for a relatively small, tightly focused team to develop a clinically relevant, technically sound AI system by designing systems transparently, with modularity and end-user needs in mind. Documented contributions here validated segmentation pipelines, PCA measurement, atlas based localization, dice derived uncertainty and a guardrail enforced app stack all bring the discipline closer to trustworthy AI tools that refine rather than distort clinical judgment.

Health care represents one of the areas in which AI has the greatest potential to affect human well-being: shortening the interval between symptom and diagnosis, providing standardization of measures which is currently undermined by variability, and bringing specialist-level analysis to where specialists are in short supply. CARVIS is one such step in this process. The work presented here is merely a beginning, not the end, and the next steps outlined above show a clear route toward an implementable system in routine clinical practice.